

Module Review: Bacterial DNA-directed RNA polymerase core enzyme in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Bucket:** kegg:ppu03020 "RNA polymerase" (module area: other_kegg_pathway) **Module scope:** bacterial core enzyme $\alpha_2\beta\beta'\omega$ (excludes sigma factors, elongation factors, promoter-specific regulators) **Overall verdict: Module fully COVERED — high confidence.** All four core subunits are present, single-copy, structurally intact, and syntenically corroborated.

1. Executive summary

The bacterial DNA-directed RNA polymerase (RNAP) core enzyme module is **completely satisfiable** in *P. putida* KT2440. All four expected steps map to unambiguous, high-confidence candidate genes:

Module step	Gene	Locus	UniProt	KO	Length	Verdict
α -dimer assembly platform	rpoA	PP_0479	Q88QL1	K03040	333 aa	covered
β/β' catalytic cleft	rpoB	PP_0447	Q88QP2	K03043	1357 aa	covered
β/β' catalytic cleft	rpoC	PP_0448	Q88QP1	K03046	1399 aa	covered
ω assembly/stability	rpoZ	PP_5301	Q88C82	K03060	87 aa	covered

Evidence is convergent across four independent lines: (i) canonical KEGG/UniProt orthology; (ii) complete, non-truncated Pfam/InterPro domain architectures including the intact catalytic **NADFDGD** Mg^{2+} -triad in rpoC; (iii) strict single-copy status with no rpoC1/rpoC2 split; (iv) conserved operon synteny (β -operon, α -operon, rpoZ–spoT linkage). The only curation caveats are annotation-hygiene issues, not gaps: EC 2.7.7.6 / GO:0003899 is propagated as a complex-

level attribute onto the non-catalytic subunits rpoA and rpoZ, and all four genes rest on homology-level evidence (protein-existence level 3) rather than direct KT2440 biochemistry. No module revision or new GO term is required.

2. Target-organism pathway definition

Process included: DNA-templated RNA synthesis by the bacterial RNAP **core enzyme** — the minimal catalytically competent assembly $\alpha_2\beta\beta'\omega$ that polymerizes ribonucleotides on a DNA template (EC 2.7.7.6). The α -dimer nucleates assembly; β and β' form the catalytic cleft housing the Mg^{2+} active center and the main/secondary channels; ω clamps β' and stabilizes the assembled enzyme.

Boundaries — keep separate: - **Sigma factors** (promoter recognition; convert core → holoenzyme): rpoD/ σ^{70} (PP_0387, K03086), rpoN/ σ^{54} (PP_0952, K03092), rpoS/ σ^{38} (PP_1623, K03087), plus other ECF/alternative sigmas. These are *correctly excluded* from this bucket. - **Elongation/anti-termination and cleavage factors** (GreA/B, NusA/G, Mfd, Rho) — separate. - **Promoter-specific transcriptional regulators / stringent-response effectors** (DksA, RelA/SpoT) — separate, though rpoZ is genomically linked to spoT (see §5). - **KEGG map03020** is a *pan-domain* reference map that also displays eukaryotic Pol I/II/III and archaeal RNAP subunits; only the four bacterial core KOs above are relevant to this organism. Do not import eukaryotic/archaeal subunit rows.

Alternate names / database definitions: "RNA polymerase" (KEGG ppu03020, DBLINK GO:0000428); GO:0000428 "DNA-directed RNA polymerase complex"; the core enzyme is EC 2.7.7.6 ("DNA-directed RNA polymerase"). Subunits are also referred to by their eukaryotic-homolog Pfam clans (Rpb1= β' , Rpb2= β , Rpb3/Rpb11 $\approx\alpha$, Rpb6= ω).

3. Expected step model

1. **RpoA α -dimer platform** — two α subunits dimerize via their N-terminal domains (α NTD) to form the assembly scaffold onto which β and β' are recruited; the α C-terminal domains (α CTD) are mobile and contact UP-element DNA and many activators.
2. **RpoB/RpoC catalytic cleft** — β (Rpb2 homolog) and β' (Rpb1 homolog) assemble on the α -dimer to build the crab-claw catalytic center; β' contributes the absolutely conserved

NADFDGD aspartate triad that chelates the catalytic Mg^{2+} , and an N-terminal Zn-binding domain.

- RpoZ ω assembly/stability subunit** — ω wraps around the β' C-terminus, chaperoning β' folding and stabilizing the assembled core.

All three steps are present and intact in KT2440 (§4).

4. Candidate genes and evidence

rpoA — PP_0479 / Q88QL1 (K03040, 333 aa). α NTD dimerization domain (PF01000) **and** α CTD (PF03118) both present → full-length, assembly-competent and regulation-competent. InterPro bacterial- α signatures (IPR011773). *Role*: assembly platform (GO:0046983 protein dimerization). *Evidence*: homology + domain architecture + α -operon synteny (rpsK–rpsD–**rpoA**–rplQ). *Caveat*: EC 2.7.7.6/GO:0003899 is a complex-level attribute, not individual catalysis. Single-copy. **Verdict: covered.**

rpoB — PP_0447 / Q88QP2 (K03043, 1357 aa). Complete Rpb2 domain set (PF04563/04561/04565/10385/00562/04560); length matches *E. coli* β (1342 aa). Harbors the rifampicin-binding pocket (clinically/experimentally validated target of rpoB mutations in *Pseudomonas*;

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Role: catalytic cleft. *Evidence*: homology + domain architecture + β -operon synteny (rplJ–rplL–**rpoB**–rpoC). Single-copy. **Verdict: covered.**

rpoC — PP_0448 / Q88QP1 (K03046, 1399 aa). Complete Rpb1 domain set (PF04997/00623/04983/05000/04998); bacterial- β' InterPro (IPR012754). **Intact catalytic NADFDGD Mg^{2+} -triad at position 458** and N-terminal Zn-finger; UniProt records GO:0000287 (Mg^{2+}) and GO:0008270 (Zn^{2+}) binding. This is the strongest single indicator that the core is catalytically competent. *Evidence*: homology + motif + synteny. Single-copy; **no rpoC1/rpoC2 split. Verdict: covered.**

rpoZ — PP_5301 / Q88C82 (K03060, 87 aa). Clean Rpb6/omega domain (PF01192, IPR006110); length matches *E. coli* ω (91 aa). *Role*: assembly/stability chaperone of β' . *Evidence*: homology + domain + synteny (directly upstream of **spoT** PP_5302). Single-copy. *Caveat*: dual-function annotation — the assembly/stability role transfers strongly, but any ppGpp/stringent-response function is an **uncertain species transfer** (see §5). **Verdict: covered.**

Evidence type for all four: UniProt protein-existence level 3 ("Inferred from homology") — i.e., no published KT2440-specific protein-level biochemistry; annotations derive from HAMAP/UniRule orthology. This is expected and unproblematic for universal housekeeping genes, but it means direct target-organism experimental support is absent. **Organism-specific genetic support:** a genome-wide RB-TnSeq library in KT2440 disrupted "nearly all (4,778) non-essential genes"

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operationally defining an essential-gene complement into which the obligatory RNAP core subunits fall — indirect but strain-specific evidence of indispensability that complements the homology annotations.

5. Gaps, ambiguities, and likely over-annotations

- **No true gaps.** Every module step is covered; no step is "missing but present under a different name." There is no lineage-specific replacement: *Pseudomonas* does not split *rpoC* (KEGG K13797/K13798 = 0 hits), unlike some cyanobacteria/*Helicobacter*.
- **Correctly absent (not expected in target taxon):** the Firmicutes/Gram-positive δ subunit (*rpoE*, K03048) has 0 hits in KT2440 — as expected for a Proteobacterium, whose core is $\alpha_2\beta\beta'\omega$ with no δ . Do not add a δ -subunit step for this organism.
- **Over-propagated catalytic annotation (benign):** EC 2.7.7.6 and molecular-function GO:0003899 ("DNA-directed RNA polymerase activity") are attached to **rpoA and rpoZ**, which have no independent nucleotidyl-transferase activity. This is a holoenzyme attribute mechanically inherited by every subunit. Prefer expressing it via complex membership (GO:0000428, `part_of`) or the `contributes_to` qualifier at the gene level.
- **Omega functional caveat (species transfer):** In *E. coli*, ω facilitates ppGpp binding and stringent-response control; but "key residues that facilitate ppGpp binding by ω are not conserved in *S. aureus*" and *rpoZ* deletion there caused "increased degradation and misfolding of the β' subunit ... and a general dissociation of RNAP"

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Thus the **assembly/stability** role is a strong transfer to KT2440; the **ppGpp-binding** role is uncertain and should not be asserted for PP_5301 without direct evidence. The genomic *rpoZ*–*spoT* adjacency is suggestive but not proof.

- **Homology-only evidence:** all four are protein-existence level 3; a skeptical reviewer would note there is no KT2440 knockout/biochemistry cited, only orthology + essentiality expectation.

6. Module and GO-curation recommendations

Module step	Status	Rationale
RpoA α -dimer platform	covered	rpoA/PP_0479, α NTD+ α CTD intact, single-copy
RpoB/RpoC catalytic cleft	covered	rpoB+rpoC, full domain sets, NADFDGD triad intact
RpoZ ω assembly/stability	covered	rpoZ/PP_5301, Rpb6 domain, single-copy

- **Do not** mark any of the three module steps (α -platform, β/β' cleft, ω) as `candidate_uncertain`, `gap`, or `not_expected_in_target_taxon` — all three are covered. (Separately, the Firmicutes δ subunit is not a step in this Proteobacterial module and is correctly absent; see §5.)
- **Module boundaries are correct** for this organism — sigma factors, elongation factors, and stringent-response effectors are separate genes and rightly excluded. No `module_needs_revision`.
- **No new module document or GO-term request needed.** Existing GO terms suffice: GO:0000428 (complex), GO:0003899 (activity, complex-level), GO:0006351 (DNA-templated transcription), GO:0046983 (rpoA dimerization).
- **GO-hygiene recommendation:** for rpoA and rpoZ, annotate catalytic function with `contributes_to` / `part_of` rather than a direct GO:0003899 assertion, to avoid implying independent catalysis.

7. Genes to promote to full review

Priority is low — the module is unambiguous — but if a `fetch-gene` pass is run:

1. **rpoZ / PP_5301** (*highest value*) — smallest, most functionally variable subunit; resolve whether the *E. coli*-type ppGpp-binding residues are conserved and whether the rpoZ–spoT linkage implies a stringent-response role in KT2440. Recommended.
2. **rpoC / PP_0448** — confirm the NADFDGD active site and Zn-finger coordinates at residue level for the structural annotation record. Optional.
3. **rpoB / PP_0447** — document the rifampicin-resistance-determining region for antibiotic-resistance curation cross-links. Optional.
4. **rpoA / PP_0479** — lowest priority; annotation is unambiguous.

8. Key references

- Zhang G. *et al.* Crystal structure of *Thermus aquaticus* core RNA polymerase at 3.3 Å. **Cell** 98:811–824 (1999).

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(KEGG *ppu03020* primary reference; core $\alpha_2\beta\beta'\omega$ architecture.)

- Weiss A. *et al.* The ω subunit governs RNA polymerase stability and transcriptional specificity in *Staphylococcus aureus*. **J Bacteriol** (2017).

P 27799328

(ω assembly/stability role; non-conservation of ppGpp-binding residues — species-transfer caveat.)

- Khitiri *et al.* The ω subunit stabilizes transcribing RNA polymerase to balance processivity and collision resolution (2026).

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(Refined mechanistic role of ω .)

- Alam K. *et al.* Rifampicin-resistant (rpoB) mutation in *Pseudomonas stutzeri* ... (2022).

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(*rpoB* as validated rifampicin target in *Pseudomonas*.)

- Cai *et al.* Novel antibiotic susceptibility of an RNA polymerase α -subunit mutant in *Pseudomonas aeruginosa* (2023).

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(*rpoA* α CTD functional relevance in *Pseudomonas*.)

- Eng T. *et al.* Engineering *Pseudomonas putida* for efficient aromatic conversion ... high-throughput screening in a bioreactor (2021).

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(Genome-wide RB-TnSeq library over ~4,778 non-essential genes $\rightarrow$ defines the essential complement containing RNAP core genes.)

- UniProt: Q88QL1 (*rpoA*), Q88QP2 (*rpoB*), Q88QP1 (*rpoC*), Q88C82 (*rpoZ*). KEGG: ppu03020 and KOs K03040/K03043/K03046/K03060.

Uncertainty statement

All four gene assignments rest on strong, convergent homology/orthology, domain-architecture, copy-number, and synteny evidence, but **no direct *P. putida* KT2440 protein-level experiment** was identified (protein-existence level 3). The core-enzyme catalytic competence is inferred from the intact NADFDGD/Zn-finger motifs. Species transfer is **strong** for all structural/assembly roles; it is **uncertain** only for the specific ppGpp-binding function of ω . Experiments that would close remaining gaps: a KT2440 *rpoZ* knockout (essentiality/stringent-response phenotype), and proteomic detection of all four subunits to raise evidence above homology level.

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